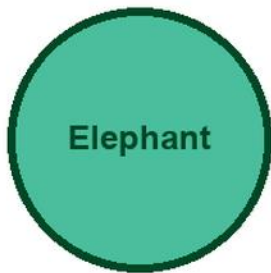


KITABU QUEST

Form I

MATHEMATICS



Elephant

Cover scene

students solving mathematics with charts and models with an elephant mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

I

Title Page

KITABU QUEST Form I Mathematics

Tanzania TIE-BASIC learner book

Mascot: elephant

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Form I learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Numbers Sets And Place Value
2. Fractions Decimals Percentages And Ratios
3. Algebraic Expressions And Linear Equations
4. Geometry Lines Angles And Polygons
5. Mensuration Of Plane Figures
6. Statistics Tables And Graphs
7. Commercial Arithmetic Basics
8. Form I Mathematics Project

As you finish each topic, return to the success criteria and correct one answer before moving on.

Numbers Sets And Place Value: Lesson Opener

Learning focus: Numbers Sets And Place Value

Country link: a Tanzania school, market, farming, building, transport, or data problem for Numbers Sets And Place Value.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about numbers sets and place value

Inquiry questions:

- How can numbers sets and place value help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Numbers Sets And Place Value: Learn

Numbers Sets And Place Value teaches you to read, write, compare, and calculate with large whole numbers.

Core idea:

- A digit has value because of its position: ones, tens, hundreds, thousands, ten-thousands, and hundred-thousands.
- Compare numbers from left to right, starting with the greatest place value.
- For addition and subtraction, align place values before calculating.
- For multiplication or division, check whether the answer should be bigger or smaller than the starting number.

Tanzania example: class enrolment, harvest records, market stock, and school budget totals can all use whole-number operations.

Key words:

- Numbers Sets And Place Value
- Numbers
- Sets
- Place
- Value
- method
- model
- unit

Numbers Sets And Place Value: Worked Example

Worked example for Numbers Sets And Place Value: A school library has 23,475 storybooks and 18,320 revision books. How many books are there altogether?

1. Align place values.
2. Add ones, tens, hundreds, thousands, and ten-thousands.
3. $23,475 + 18,320 = 41,795$.
4. Estimate: $23,000 + 18,000 = 41,000$, so 41,795 is reasonable.

Answer: 41,795 books.

Numbers Sets And Place Value: Vocabulary And Study Skill

Use these words and study moves before you practise Numbers Sets And Place Value. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Numbers Sets And Place Value
- Numbers
- Sets
- Place
- Value
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Numbers Sets And Place Value without a clear example, method, or evidence.

Numbers Sets And Place Value: Guided Activity

Guided activity for Numbers Sets And Place Value:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Numbers Sets And Place Value: Practice And Correction

Practice for Numbers Sets And Place Value:

Main outcome: solve and explain problems about numbers sets and place value.

1. Solve one numbers sets and place value question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to numbers sets and place value and solve it.

Numbers Sets And Place Value: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Numbers Sets And Place Value.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Numbers Sets And Place Value: Outcome Check

Outcome check for Numbers Sets And Place Value: Solve and explain problems about numbers sets and place value.

1. Add 34,785 and 21,406.
2. Estimate the answer to check it.
3. Explain one place-value step.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions Decimals Percentages And Ratios: Lesson Opener

Learning focus: Fractions Decimals Percentages And Ratios

Country link: a Tanzania school, market, farming, building, transport, or data problem for Fractions Decimals Percentages And Ratios.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about fractions decimals percentages and ratios

Inquiry questions:

- How can fractions decimals percentages and ratios help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions Decimals Percentages And Ratios: Learn

Fractions Decimals Percentages And Ratios teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Fractions Decimals Percentages And Ratios
- Fractions
- Decimals
- Percentages
- Ratios
- method
- model
- unit

Fractions Decimals Percentages And Ratios: Worked Example

Worked example for Fractions Decimals Percentages And Ratios: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Fractions Decimals Percentages And Ratios: Vocabulary And Study Skill

Use these words and study moves before you practise Fractions Decimals Percentages And Ratios. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Fractions Decimals Percentages And Ratios
- Fractions
- Decimals
- Percentages
- Ratios
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Fractions Decimals Percentages And Ratios without a clear example, method, or evidence.

Fractions Decimals Percentages And Ratios: Guided Activity

Guided activity for Fractions Decimals Percentages And Ratios:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Fractions Decimals Percentages And Ratios: Practice And Correction

Practice for Fractions Decimals Percentages And Ratios:

Main outcome: solve and explain problems about fractions decimals percentages and ratios.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to fractions decimals percentages and ratios and solve it.

Fractions Decimals Percentages And Ratios: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Fractions Decimals Percentages And Ratios.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions Decimals Percentages And Ratios: Outcome Check

Outcome check for Fractions Decimals Percentages And Ratios: Solve and explain problems about fractions decimals percentages and ratios.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic Expressions And Linear Equations: Lesson Opener

Learning focus: Algebraic Expressions And Linear Equations

Country link: a Tanzania school, market, farming, building, transport, or data problem for Algebraic Expressions And Linear Equations.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about algebraic expressions and linear equations

Inquiry questions:

- How can algebraic expressions and linear equations help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic Expressions And Linear Equations: Learn

Algebraic Expressions And Linear Equations teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Algebraic Expressions And Linear Equations
- Algebraic
- Expressions
- Linear
- Equations
- method
- model
- unit

Algebraic Expressions And Linear Equations: Worked Example

Worked example for Algebraic Expressions And Linear Equations: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Algebraic Expressions And Linear Equations: Vocabulary And Study Skill

Use these words and study moves before you practise Algebraic Expressions And Linear Equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Algebraic Expressions And Linear Equations
- Algebraic
- Expressions
- Linear
- Equations
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Algebraic Expressions And Linear Equations without a clear example, method, or evidence.

Algebraic Expressions And Linear Equations: Guided Activity

Guided activity for Algebraic Expressions And Linear Equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Algebraic Expressions And Linear Equations: Practice And Correction

Practice for Algebraic Expressions And Linear Equations:

Main outcome: solve and explain problems about algebraic expressions and linear equations.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to algebraic expressions and linear equations and solve it.

Algebraic Expressions And Linear Equations: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Algebraic Expressions And Linear Equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic Expressions And Linear Equations: Outcome Check

Outcome check for Algebraic Expressions And Linear Equations: Solve and explain problems about algebraic expressions and linear equations.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Geometry Lines Angles And Polygons: Lesson Opener

Learning focus: Geometry Lines Angles And Polygons

Country link: a Tanzania school, market, farming, building, transport, or data problem for Geometry Lines Angles And Polygons.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about geometry lines angles and polygons

Inquiry questions:

- How can geometry lines angles and polygons help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Geometry Lines Angles And Polygons: Learn

Geometry Lines Angles And Polygons teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Geometry Lines Angles And Polygons
- Geometry
- Lines
- Angles
- Polygons
- method
- model
- unit

Geometry Lines Angles And Polygons: Worked Example

Worked example for Geometry Lines Angles And Polygons: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Geometry Lines Angles And Polygons: Vocabulary And Study Skill

Use these words and study moves before you practise Geometry Lines Angles And Polygons. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Geometry Lines Angles And Polygons
- Geometry
- Lines
- Angles
- Polygons
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Geometry Lines Angles And Polygons without a clear example, method, or evidence.

Geometry Lines Angles And Polygons: Guided Activity

Guided activity for Geometry Lines Angles And Polygons:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Geometry Lines Angles And Polygons: Practice And Correction

Practice for Geometry Lines Angles And Polygons:

Main outcome: solve and explain problems about geometry lines angles and polygons.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to geometry lines angles and polygons and solve it.

Geometry Lines Angles And Polygons: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Geometry Lines Angles And Polygons.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Geometry Lines Angles And Polygons: Outcome Check

Outcome check for Geometry Lines Angles And Polygons: Solve and explain problems about geometry lines angles and polygons.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mensuration Of Plane Figures: Lesson Opener

Learning focus: Mensuration Of Plane Figures

Country link: a Tanzania school, market, farming, building, transport, or data problem for Mensuration Of Plane Figures.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about mensuration of plane figures

Inquiry questions:

- How can mensuration of plane figures help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mensuration Of Plane Figures: Learn

Mensuration Of Plane Figures teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Mensuration Of Plane Figures
- Mensuration
- Plane
- Figures
- method
- model
- unit
- answer

Mensuration Of Plane Figures: Worked Example

Worked example for Mensuration Of Plane Figures: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Mensuration Of Plane Figures: Vocabulary And Study Skill

Use these words and study moves before you practise Mensuration Of Plane Figures. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Mensuration Of Plane Figures
- Mensuration
- Plane
- Figures
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Mensuration Of Plane Figures without a clear example, method, or evidence.

Mensuration Of Plane Figures: Guided Activity

Guided activity for Mensuration Of Plane Figures:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Mensuration Of Plane Figures: Practice And Correction

Practice for Mensuration Of Plane Figures:

Main outcome: solve and explain problems about mensuration of plane figures.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to mensuration of plane figures and solve it.

Mensuration Of Plane Figures: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Mensuration Of Plane Figures.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Mensuration Of Plane Figures: Outcome Check

Outcome check for Mensuration Of Plane Figures: Solve and explain problems about mensuration of plane figures.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics Tables And Graphs: Lesson Opener

Learning focus: Statistics Tables And Graphs

Country link: a Tanzania school, market, farming, building, transport, or data problem for Statistics Tables And Graphs.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about statistics tables and graphs

Inquiry questions:

- How can statistics tables and graphs help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics Tables And Graphs: Learn

Statistics Tables And Graphs teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Statistics Tables And Graphs
- Statistics
- Tables
- Graphs
- method
- model
- unit
- answer

Statistics Tables And Graphs: Worked Example

Worked example for Statistics Tables And Graphs: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Statistics Tables And Graphs: Vocabulary And Study Skill

Use these words and study moves before you practise Statistics Tables And Graphs. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Statistics Tables And Graphs
- Statistics
- Tables
- Graphs
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Statistics Tables And Graphs without a clear example, method, or evidence.

Statistics Tables And Graphs: Guided Activity

Guided activity for Statistics Tables And Graphs:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Statistics Tables And Graphs: Practice And Correction

Practice for Statistics Tables And Graphs:

Main outcome: solve and explain problems about statistics tables and graphs.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to statistics tables and graphs and solve it.

Statistics Tables And Graphs: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Statistics Tables And Graphs.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics Tables And Graphs: Outcome Check

Outcome check for Statistics Tables And Graphs: Solve and explain problems about statistics tables and graphs.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Commercial Arithmetic Basics: Lesson Opener

Learning focus: Commercial Arithmetic Basics

Country link: a Tanzania school, market, farming, building, transport, or data problem for Commercial Arithmetic Basics.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about commercial arithmetic basics

Inquiry questions:

- How can commercial arithmetic basics help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Commercial Arithmetic Basics: Learn

Commercial Arithmetic Basics teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Commercial Arithmetic Basics
- Commercial
- Arithmetic
- Basics
- method
- model
- unit
- answer

Commercial Arithmetic Basics: Worked Example

Worked example for Commercial Arithmetic Basics: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Tanzania classroom, market, garden, transport, or home example.

Commercial Arithmetic Basics: Vocabulary And Study Skill

Use these words and study moves before you practise Commercial Arithmetic Basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Commercial Arithmetic Basics
- Commercial
- Arithmetic
- Basics
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Commercial Arithmetic Basics without a clear example, method, or evidence.

Commercial Arithmetic Basics: Guided Activity

Guided activity for Commercial Arithmetic Basics:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Commercial Arithmetic Basics: Practice And Correction

Practice for Commercial Arithmetic Basics:

Main outcome: solve and explain problems about commercial arithmetic basics.

1. Solve one commercial arithmetic basics question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to commercial arithmetic basics and solve it.

Commercial Arithmetic Basics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Commercial Arithmetic Basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Commercial Arithmetic Basics: Outcome Check

Outcome check for Commercial Arithmetic Basics: Solve and explain problems about commercial arithmetic basics.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Form I Mathematics Project: Lesson Opener

Learning focus: Form I Mathematics Project

Country link: a Tanzania school, market, farming, building, transport, or data problem for Form I Mathematics Project.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve and explain problems about form i mathematics project

Inquiry questions:

- How can form i mathematics project help you solve a real problem?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Form I Mathematics Project: Learn

Form I Mathematics Project teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Form I Mathematics Project
- Form
- Mathematics
- Project
- method
- model
- unit
- answer

Form I Mathematics Project: Worked Example

Worked example for Form I Mathematics Project: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Tanzania classroom, market, garden, transport, or home example.

Form I Mathematics Project: Vocabulary And Study Skill

Use these words and study moves before you practise Form I Mathematics Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Form I Mathematics Project
- Form
- Mathematics
- Project
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form I Mathematics Project without a clear example, method, or evidence.

Form I Mathematics Project: Guided Activity

Guided activity for Form I Mathematics Project:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Form I Mathematics Project: Practice And Correction

Practice for Form I Mathematics Project:

Main outcome: solve and explain problems about form i mathematics project.

1. Solve one form i mathematics project question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to form i mathematics project and solve it.

Form I Mathematics Project: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form I Mathematics Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Form I Mathematics Project: Outcome Check

Outcome check for Form I Mathematics Project: Solve and explain problems about form i mathematics project.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Numbers Sets And Place Value: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Numbers Sets And Place Value that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Fractions Decimals Percentages And Ratios: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Fractions Decimals Percentages And Ratios that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Algebraic Expressions And Linear Equations: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Algebraic Expressions And Linear Equations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Geometry Lines Angles And Polygons: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Geometry Lines Angles And Polygons that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Numbers Sets And Place Value: Application Workshop 1

Application workshop 1 for Numbers Sets And Place Value.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Fractions Decimals Percentages And Ratios: Application Workshop 2

Application workshop 2 for Fractions Decimals Percentages And Ratios.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Algebraic Expressions And Linear Equations: Application Workshop 3

Application workshop 3 for Algebraic Expressions And Linear Equations.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Geometry Lines Angles And Polygons: Application Workshop 4

Application workshop 4 for Geometry Lines Angles And Polygons.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Mensuration Of Plane Figures: Application Workshop 5

Application workshop 5 for Mensuration Of Plane Figures.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Statistics Tables And Graphs: Application Workshop 6

Application workshop 6 for Statistics Tables And Graphs.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Commercial Arithmetic Basics: Application Workshop 7

Application workshop 7 for Commercial Arithmetic Basics.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Form I Mathematics Project: Application Workshop 8

Application workshop 8 for Form I Mathematics Project.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Numbers Sets And Place Value: Application Workshop 9

Application workshop 9 for Numbers Sets And Place Value.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Fractions Decimals Percentages And Ratios: Application Workshop 10

Application workshop 10 for Fractions Decimals Percentages And Ratios.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Glossary

Use this glossary to revise important words in Mathematics.

- Numbers Sets And Place Value: Explain this term using an example from the book, your school, or your community.
- Fractions Decimals Percentages And Ratios: Explain this term using an example from the book, your school, or your community.
- Algebraic Expressions And Linear Equations: Explain this term using an example from the book, your school, or your community.
- Geometry Lines Angles And Polygons: Explain this term using an example from the book, your school, or your community.
- Mensuration Of Plane Figures: Explain this term using an example from the book, your school, or your community.
- Statistics Tables And Graphs: Explain this term using an example from the book, your school, or your community.
- Commercial Arithmetic Basics: Explain this term using an example from the book, your school, or your community.
- Form I Mathematics Project: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.